

Unit 1 : Environmental Studies



COLLEGE OF ENGINEERING



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Lesson Outline

- ☐ **Definition and concept of environment**
- ☐ **Types of environment**
- ☐ **Scope of environmental studies**
- ☐ **Multidisciplinary nature of Environmental Studies**
- ☐ **Importance of Environmental Studies**

General Objectives

- ☐ Understand the way natural ecosystems function.
- ☐ Appreciate the ways in which human populations disrupt the functioning of the natural ecosystem and thus cause environmental problems.
- ☐ Propose and evaluate solutions to current environmental dilemmas.

Introduction to Environmental Studies



What is Environment ?

- The word environment is derived from the French word *environ*, meaning our surroundings or external conditions.



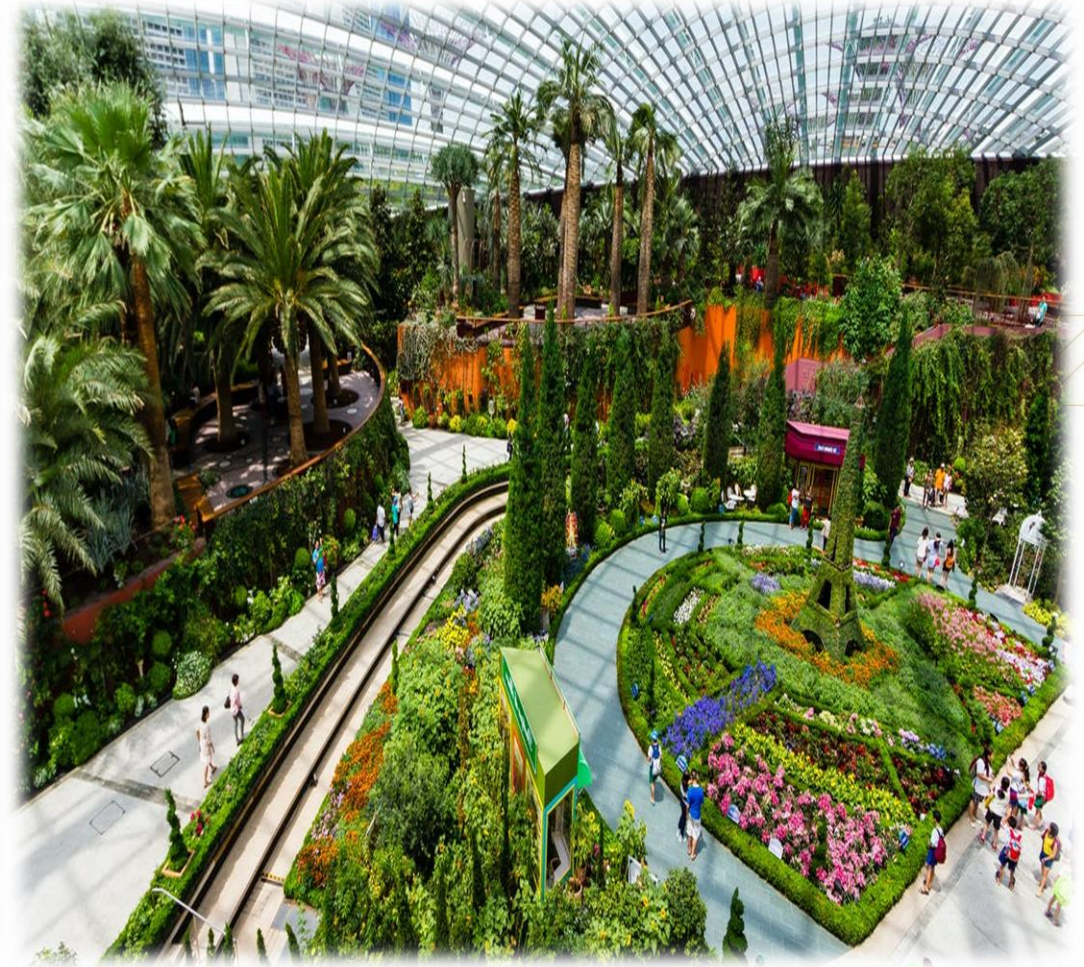
- Environment is defined as the surroundings conditions (physical, biological, social, cultural, geographical, etc) that affects the life, nature, behaviour, growth and development of Living organisms



Types of Environment



Natural Environment



Artificial/ Man made

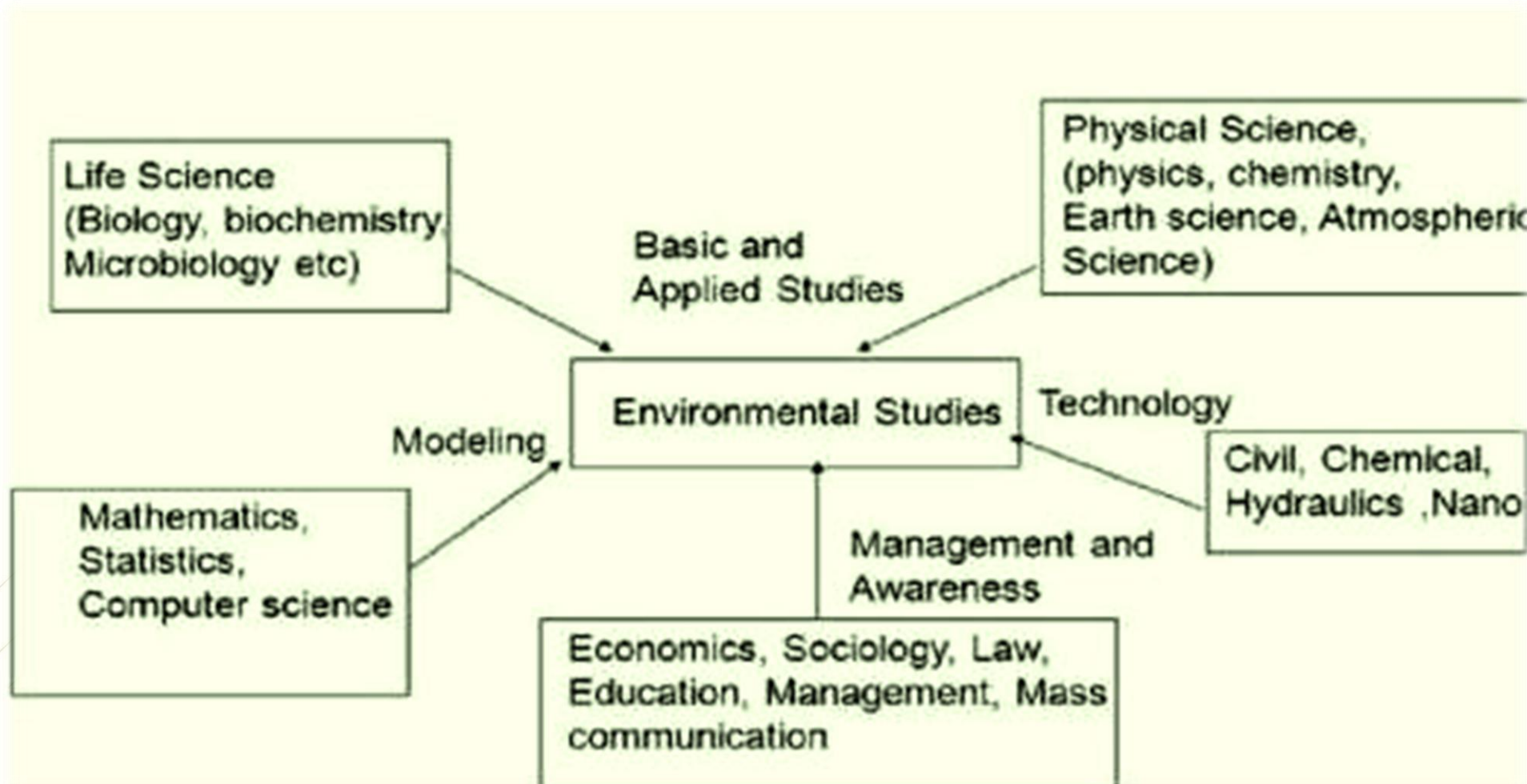
Scope of Environmental Science

Study of components, ethics, concepts and problems associated with, Pollution, natural resource depletion, population explosion and other environmental problem,



For finding practical solutions leading to effective management practices, protection and people's participation at all levels.

Multidisciplinary Nature



Importance of Environmental studies

- ❑ Environmental studies teaches people to understand their role in the environment and learn to live with limited natural resources so as to avoid future disasters**
- ❑ Proper education and public awareness to tackle environmental problems.**
- ❑ Environmental studies provide sufficient knowledge about the philosophy, genesis and consequences of local and global environmental problems and the necessary knowledge for their abatement and control**

Need for Public Awareness

- Since our environment is getting degraded due to human activities, we need to do something about it to sustain the quality.
- Environmental protection requires both preventive and curative measures.
- Government alone cannot perform all the clean-up functions. Individuals/group efforts in their own every possible way has to be made to protect our environment.
- Indian government has taken some important steps towards environmental protection . E.g. no development work can progress without a proper Environmental Impact Assessment (EIA)
- Public awareness needs to be created. Both print media and electronic media can strongly influence public opinion.
- Social awareness can encourage people to use biogas and solar energy and non conventional energy instead of fossil fuels.
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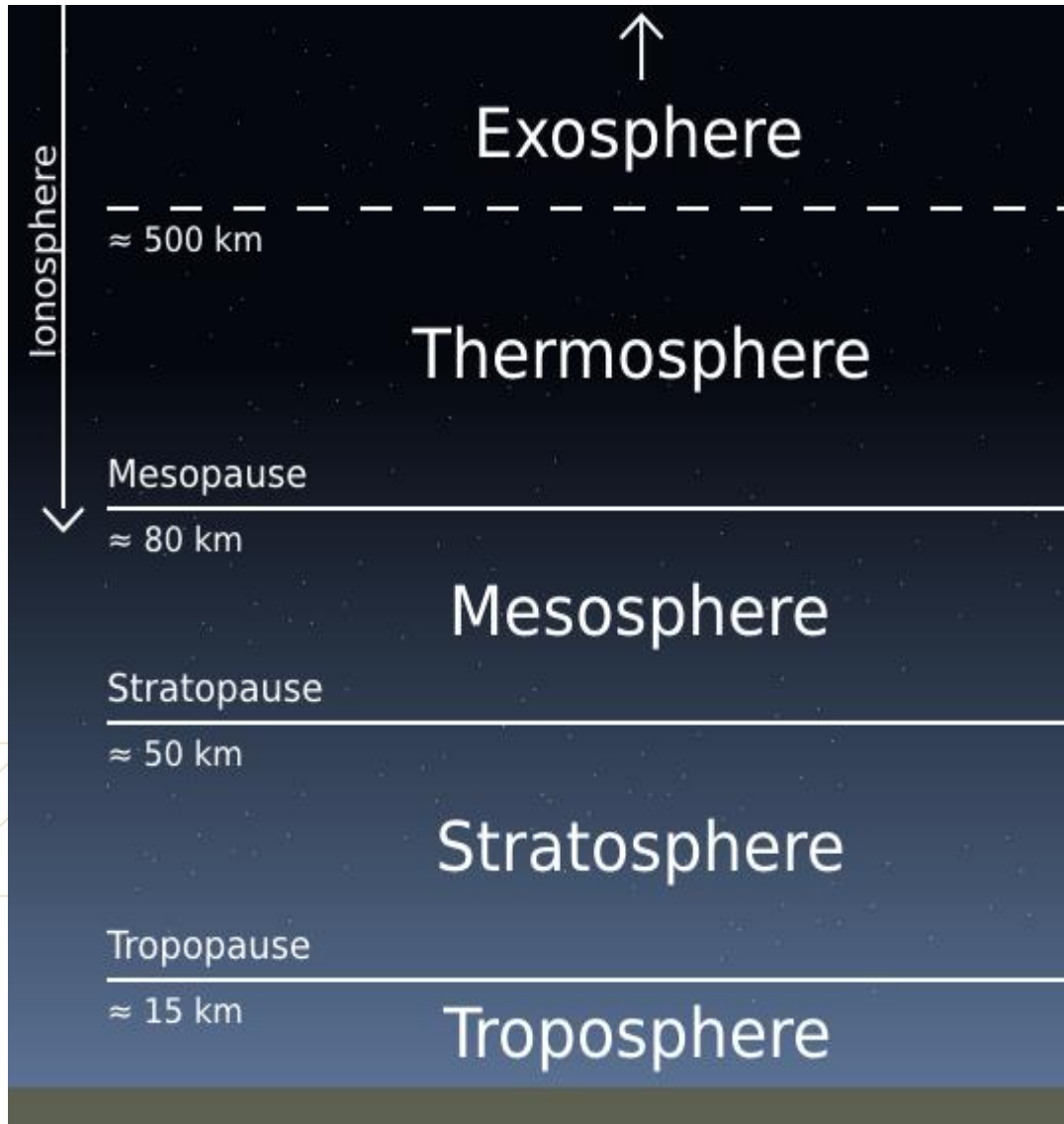
Continuation.....

- Industrial development, along with urbanization, can generate health problems. Awareness about healthy living conditions and the consequences of ill health will make people conscious about their health and hygienic living conditions..
- Environment is an integration of both living and non-living organisms. Water, air, soil, minerals, wild life, grass lands, forests, oceans, agriculture are all life supporting systems. Since these natural resources are limited, and human activities are the causative factors for environmental degradation, each one of us need to feel responsible to protect the environment

❖ Earth and its environment have the following three basic components which are:

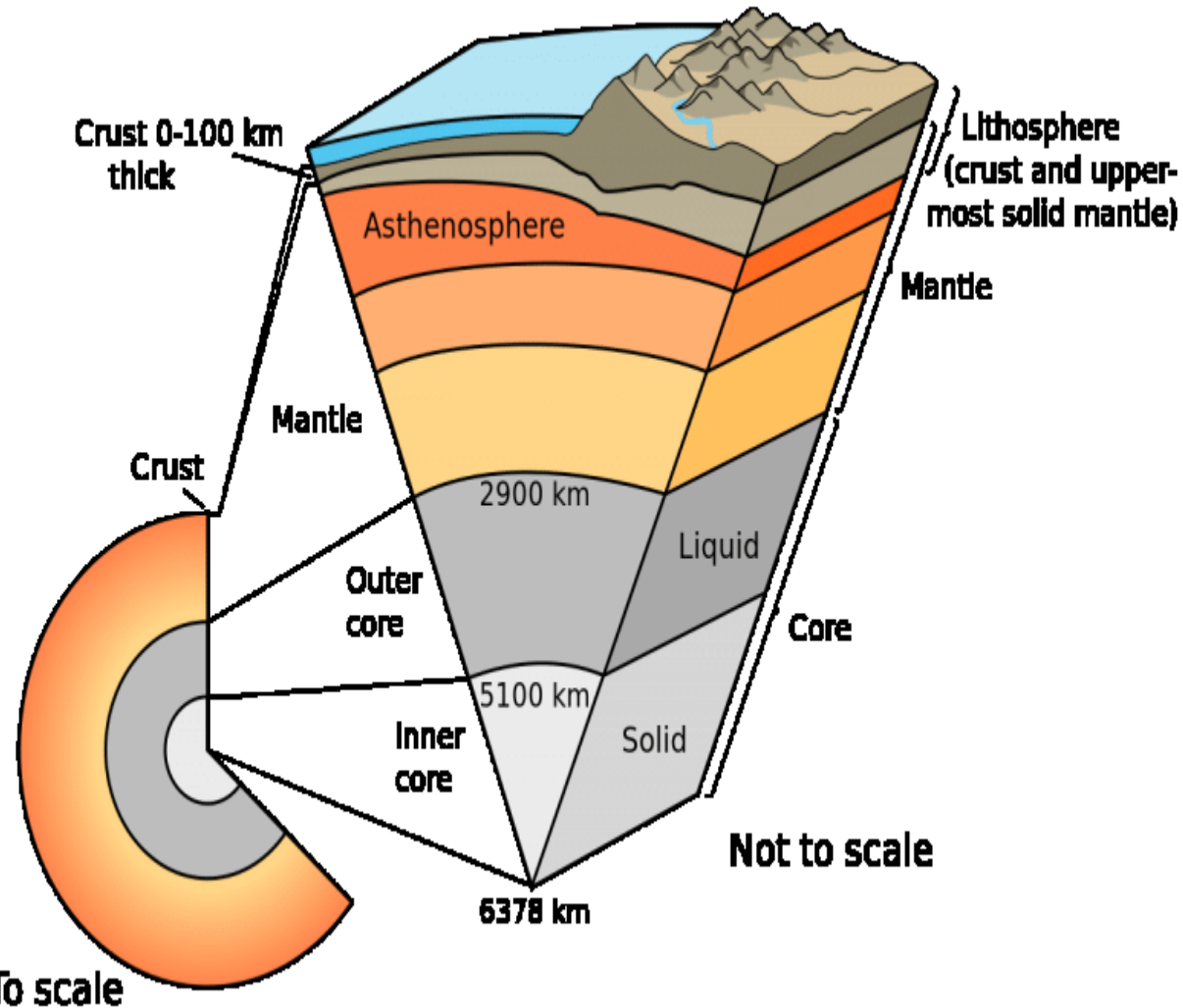
1. The atmosphere or air
2. The hydrosphere or water
3. The lithosphere or the rocks and the soil.

Structure of Atmosphere



- **Exosphere (upto 1600 km)**
- **Thermosphere/Ionosphere(85-500 km)**
- **Mesosphere (50-85 km)**
- **Stratosphere (15-50 km)**
- **Troposphere (0-15 km)**

Lithosphere



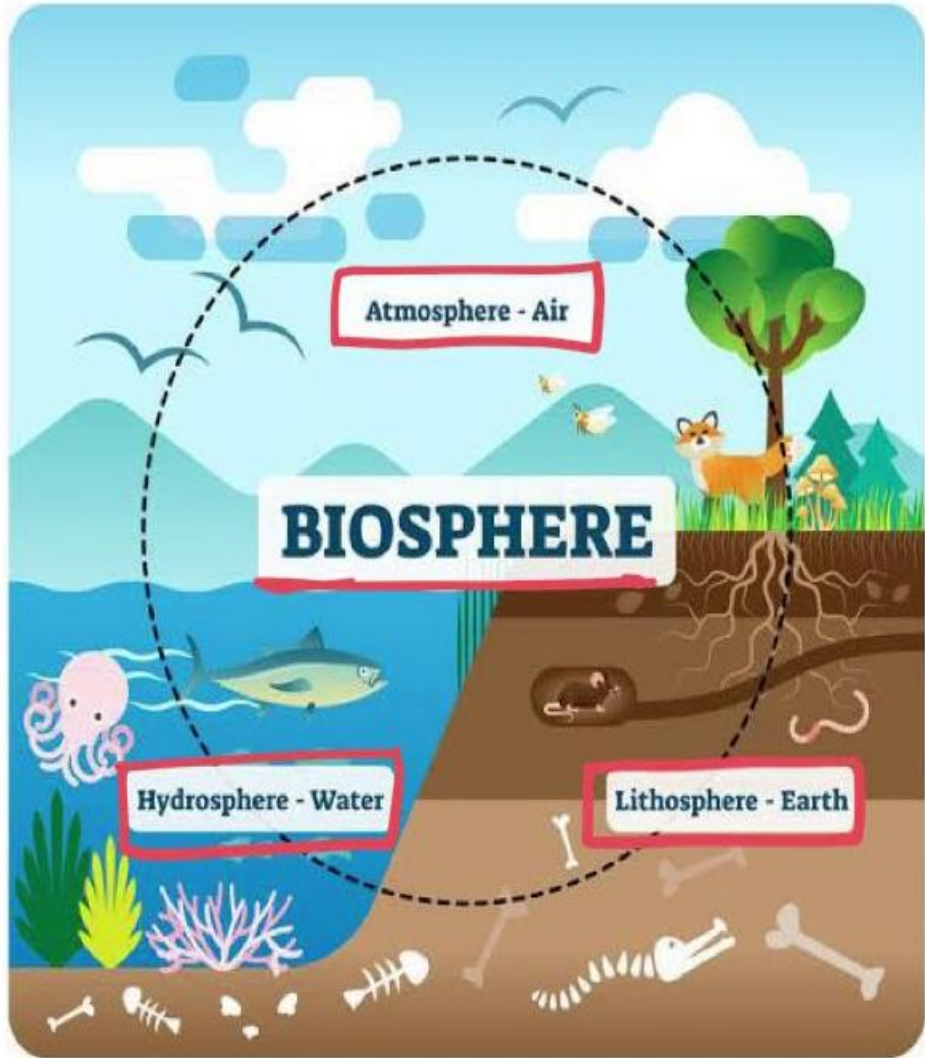
- ❑ Lithosphere is the outermost mantle of the rocks constituting the earth's crust.
- ❑ The lithosphere has a thickness ranging from 64 to 96 km. The uppermost part of the Lithosphere rich in Si and Al and is known as SiAl layer.
- ❑ The zone next to the SiAl is rich in Si and Mg. This layer is formed of basalt rocks and constitutes the ocean floors.
- ❑ Below the lithosphere lies the mantle, thickness of about 2400 km.
- ❑ The upper part of the mantle is known as the Asthenosphere, while the lower mantle is called the Mesosphere..
- ❑ The inner most part of the earth is called Core, thickness extends about 3500 km. The core consists of the outer core and the inner core.



All the water on or near Earth's Surface

- Oceans
- Lakes
- Rivers
- Wetlands
- Icecaps and Icesheets
- Soil
- Groundwater
- Clouds and atmosphere

Biosphere



- ❑ Earth has different components – Lithosphere, Atmosphere and hydrosphere
- ❑ Biosphere is that part of lithosphere, hydrosphere and atmosphere where plants and human beings live.
- ❑ It is the part of the earth where life exists
- ❑ It consists the living organisms and the dead organic matter.

THANK YOU

